

Shahzaib Saqib Warraich

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I am interested in investigating the behavioral mechanisms underlying language model reliability: how pre-training shapes factual knowledge representation, how post-training affects uncertainty calibration, and how emergent behaviors during generation reveal models' epistemic states and enable appropriate abstention.

EDUCATION

University of Southern California

Jan 2024 - Present

M.S. in Applied Data Science

National University of Sciences and Technology

Sep 2017 - June 2021

B.E. in Electrical Engineering and Computer Science

PUBLICATIONS

Sample, Align, Synthesize: Graph-Based Response Synthesis with ConGrs ([URL](#))

S Ghosh, [SS Warraich](#), D Tarsadiya, G Yauney, S Swayamdipta

ArXiv 2025

How Reliable is Language Model Micro-Benchmarking? ([URL](#))

G Yauney, [SS Warraich](#), S Swayamdipta

ArXiv 2025

RESEARCH EXPERIENCE

Sample, Align, Synthesize: Graph-Based Response Synthesis with ConGrs *Supervisor: Prof. Swabha Swayamdipta — Submitted for peer review*

- Developed and validated application of consensus decoding framework to factuality-critical tasks, conducting systematic evaluations across long-form biography generation and PopQA benchmarks with five independent replications per model, establishing robust baseline performance metrics and characterizing the factuality-information tradeoff
- Designed and implemented high-throughput batch processing infrastructure for FActScore evaluation, engineering performance optimizations including response caching that resolved critical processing bottlenecks and improved evaluation efficiency
- Architected comprehensive ablation studies investigating scaling dynamics of sample size during alignment and response synthesis phases, along with temperature-based sensitivity analysis, providing empirical insights into framework robustness
- Extended framework to refusal-aware settings across three challenging domains (Numerical False Presuppositions, Scientific Attributions, Historical Events), integrating sophisticated LLM-as-a-judge abstention detection for more nuanced assessment

How Reliable is Language Model Micro-Benchmarking? *Supervisor: Prof. Swabha Swayamdipta — Submitted for peer review*

- Proposed and implemented confidence-based stratified sampling methodology for micro-benchmark selection in meta-evaluation frameworks, adapting principled statistical techniques from computer vision [literature](#) to language model evaluation

- Developed comprehensive evaluation infrastructure integrating multiple micro-benchmark selection approaches including stratified sampling, TinyBenchmarks (P-IRT and GP-IRT), and task reduction [baselines](#), enabling systematic comparison of benchmark selection strategies
- Maintained and extended research codebase, implementing modular components for reproducible meta-evaluation experiments across diverse benchmark selection methodologies

INDUSTRY EXPERIENCE

Paramount, Data Scientist Intern May 2025 - Oct 2025
Entertainment Analytics & AI

- Enhanced content performance forecasting accuracy by up to **20%** across critical business KPIs through strategic integration of multi-platform social media engagement signals into predictive modeling frameworks
- Architected a production-grade GenAI explainability system leveraging ensemble LLM-as-a-judge methodology with majority voting consensus, enabling granular title-level performance attribution and democratizing advanced ML insights for non-technical stakeholders across the organization

Retrocausal, Research Engineer II (AI) Dec 2022 - Dec 2023
Computer Vision & Manufacturing Intelligence

- Designed and deployed precision assembly verification solutions achieving **98%** detection accuracy for Fortune 500 manufacturing clients including Schneider Electric, Carrier, and Honda through state-of-the-art object detection and activity recognition algorithms ([Assembly Copilot](#))
- Architected an automated model training infrastructure incorporating advanced cut-paste-mix augmentation techniques to generate photorealistic synthetic datasets, reducing manual annotation overhead by **80%** while maintaining rigorous performance standards
- Conceptualized and led development of a computer vision-enabled ergonomics risk assessment module for Kaizen Copilot, establishing a differentiated product capability addressing workplace safety optimization and regulatory compliance ([Kaizen Copilot](#))

Adlytic AI, AI Engineer June 2021 - Dec 2022
Edge AI & Recommender Systems

- Spearheaded comprehensive model optimization initiatives, implementing knowledge distillation and NVIDIA TensorRT optimization to achieve **66%** reduction in edge inference latency while maintaining performance within **1.5%** of baseline accuracy across production deployments
- Architected and operationalized a scalable cloud-based recommender system achieving **97%** prediction accuracy, driving a **3x** increase in product conversion rates and generating measurable revenue impact through enhanced customer engagement

AWARDS & HONORS

- USC Andrew J. Viterbi Scholarship 2024 - 2025
- William J. Fulbright Scholarship 2024 - 2025

TEACHING

Course Producer, *University of Southern California – Viterbi School of Engineering*

- CSCI 544: Applied Natural Language Processing Aug 2024 - Dec 2024
- CSCI 544: Applied Natural Language Processing Jan 2025 - May 2025

LEADERSHIP

Viterbi Graduate Student Association (VGSA), Vice President May 2025 - Present

- Led student advocacy initiatives representing 6,000 graduate students as primary liaison between student body and university administration
- Managed strategic partnerships with internal stakeholders and external organizations to enhance graduate student resources and professional opportunities
- Delivered monthly progress reports to Viterbi School Dean, communicating student concerns and actionable recommendations

SKILLS & INTERESTS

Skills

- Languages: English, French
- Programming Languages: Python, C/C++, SQL
- Deep Learning: PyTorch, Hugging Face, vLLM
- Machine Learning: Scikit-Learn, OpenCV
- Data Analysis: Pandas, Dask, PySpark
- Data Visualization: Matplotlib, Seaborn, Plotly
- Cloud: AWS Sagemaker, GCP Vertex AI
- Others: Docker, Kubernetes, MongoDB, Git, LaTeX, Figma

Interests

- AI Ethics and Responsibility
- Hackathons
- User Experience (UX) Design